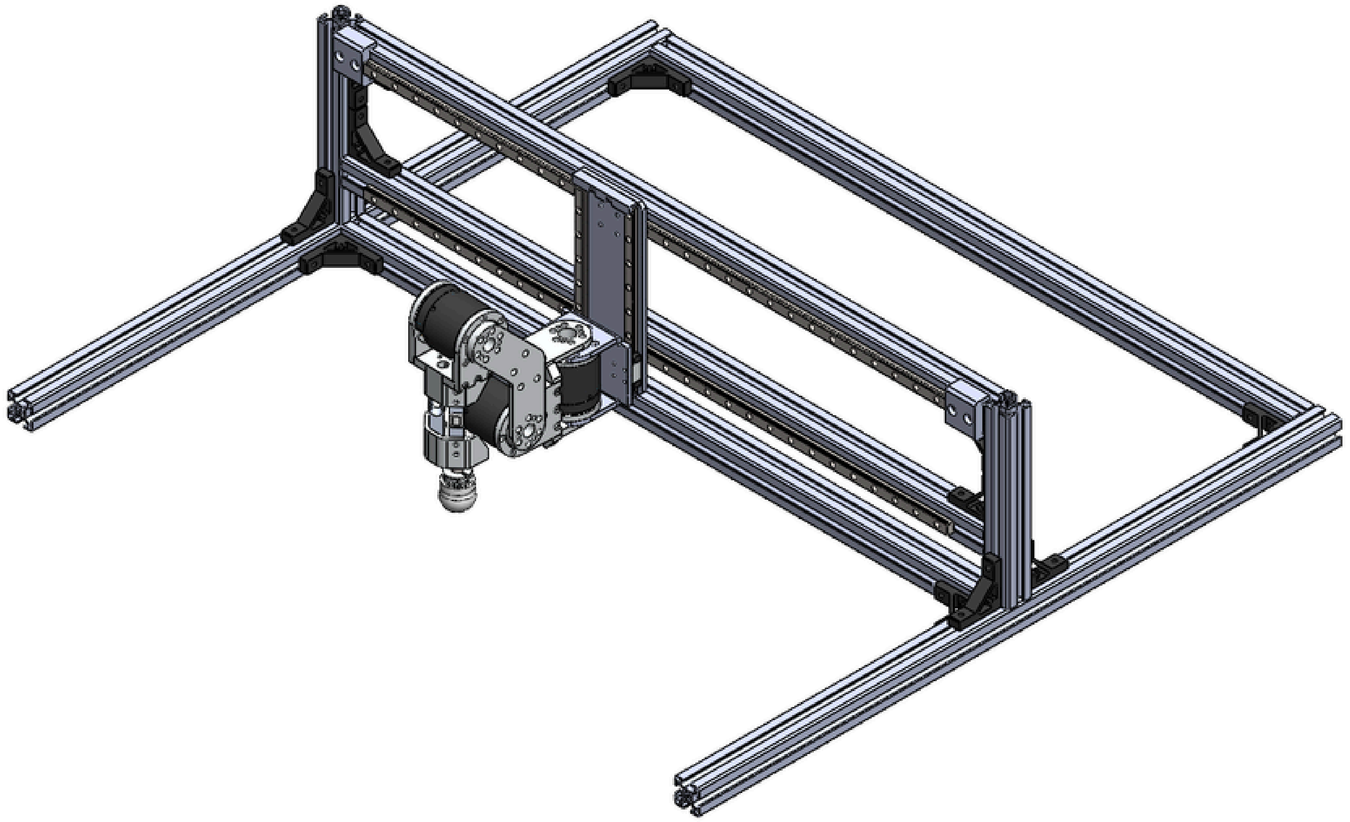
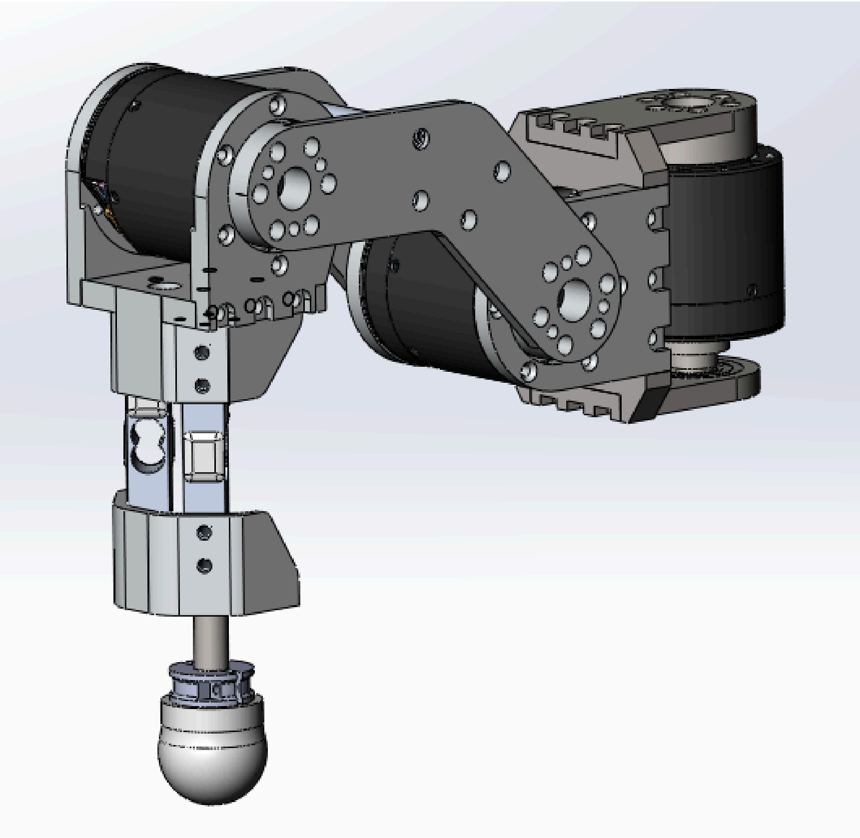




KMUTT

FIBO

สถาบันวิทยาการหุ่นยนต์ภาคสนาม (FIBO)
Institute of field Robotics

BRIDGING THE SIM-TO-REAL GAP IN HEXAPOD LOCOMOTION VIA UNSUPERVISED ACTUATOR NETWORKS

6606 6671 6674

PROBLEM

Problem: Ideal Simulation vs Real Actuators

Ideal simulator assumption

- Motor follows commands instantly.
- Friction and backlash are simplified.
- Delay and hysteresis are often missing.

Real actuator behavior

- Static friction and sliding friction.
- Gear backlash and compliance.
- Communication/control latency.
- Load and contact-dependent response.

Why this matters

If the actuator model is wrong, a DRL policy can exploit simulator behavior that does not exist on the real robot.

SCOPE

Project Scope

Original motivation

Reduce the sim-to-real gap for hexapod locomotion by improving actuator realism.

Final implementation scope

A controlled 3-DOF one-leg proof of concept:

- joints: hip, knee, ankle
- data source: real hardware CSV logs
- goal: replay hardware joint transitions more accurately in simulation

Correct interpretation

This project does not claim full hexapod walking yet. It validates a learned actuator correction layer on one leg.

Project Scope

Stage	Input/Output	Purpose
Hardware logging	commands, q, $\dot{q}$, optional τ	Capture real actuator response.
Dataset loader	CSV $\rightarrow$ tensors	Build valid replay transitions.
Isaac Lab replay	recorded command $+$ current sim state	Recreate hardware trajectory in simulation.
UAN policy	error history $\rightarrow$ residual effort	Correct actuator mismatch.
Evaluation	baseline vs UAN metrics	Measure sim-to-real replay error.

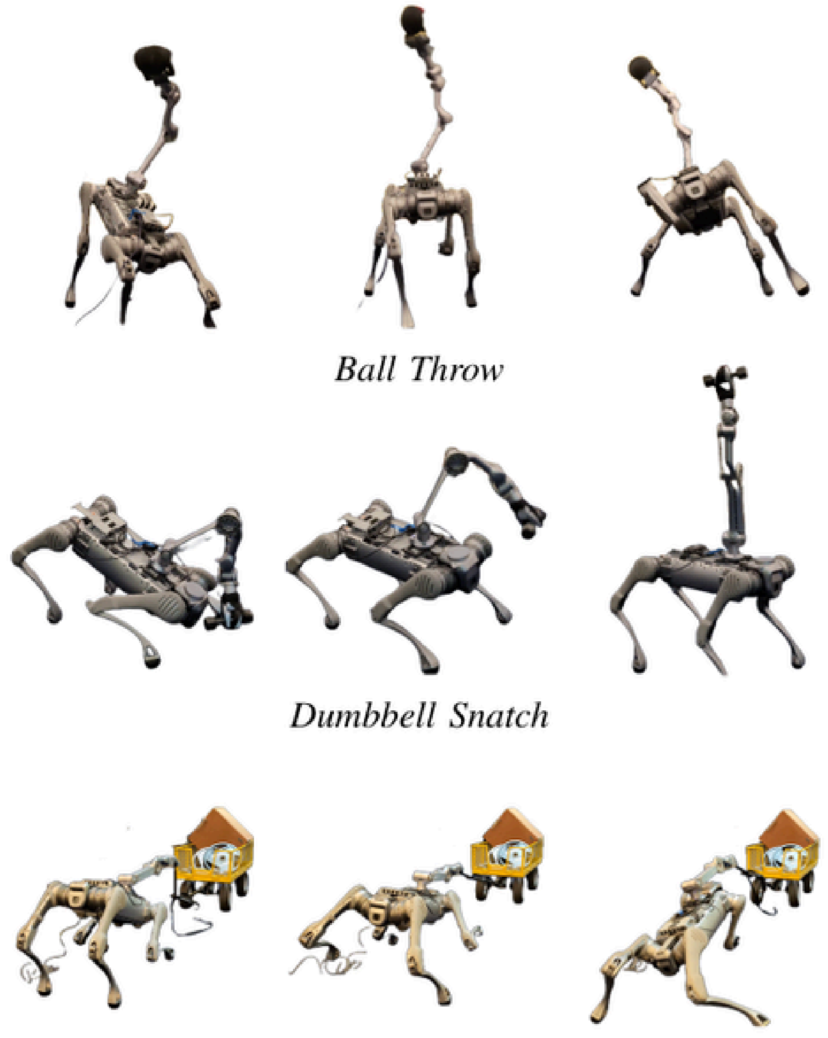
Baseline

`baseline_zero_residual`: same simulator and recorded command, but no UAN residual effort.

PAPER REVIEW

Bridging the Sim-to-Real Gap for Athletic Loco-Manipulation

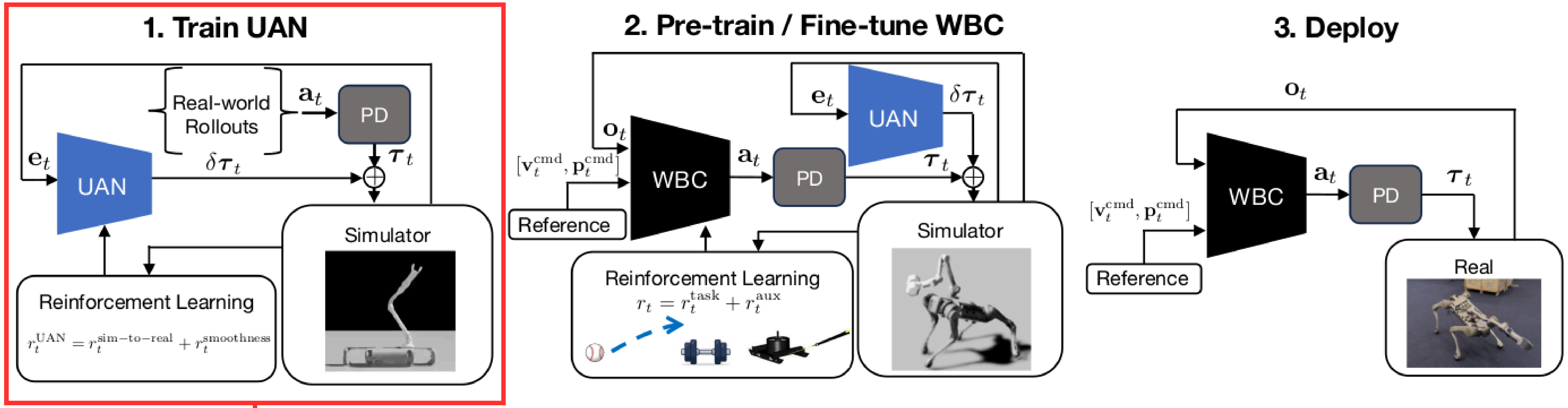
Nolan Fey, Gabriel B. Margolis, Martin Peticco, and Pulkit Agrawal
Improbable AI Lab
Massachusetts Institute of Technology, Cambridge, MA 02139



Ball Throw

Dumbbell Snatch

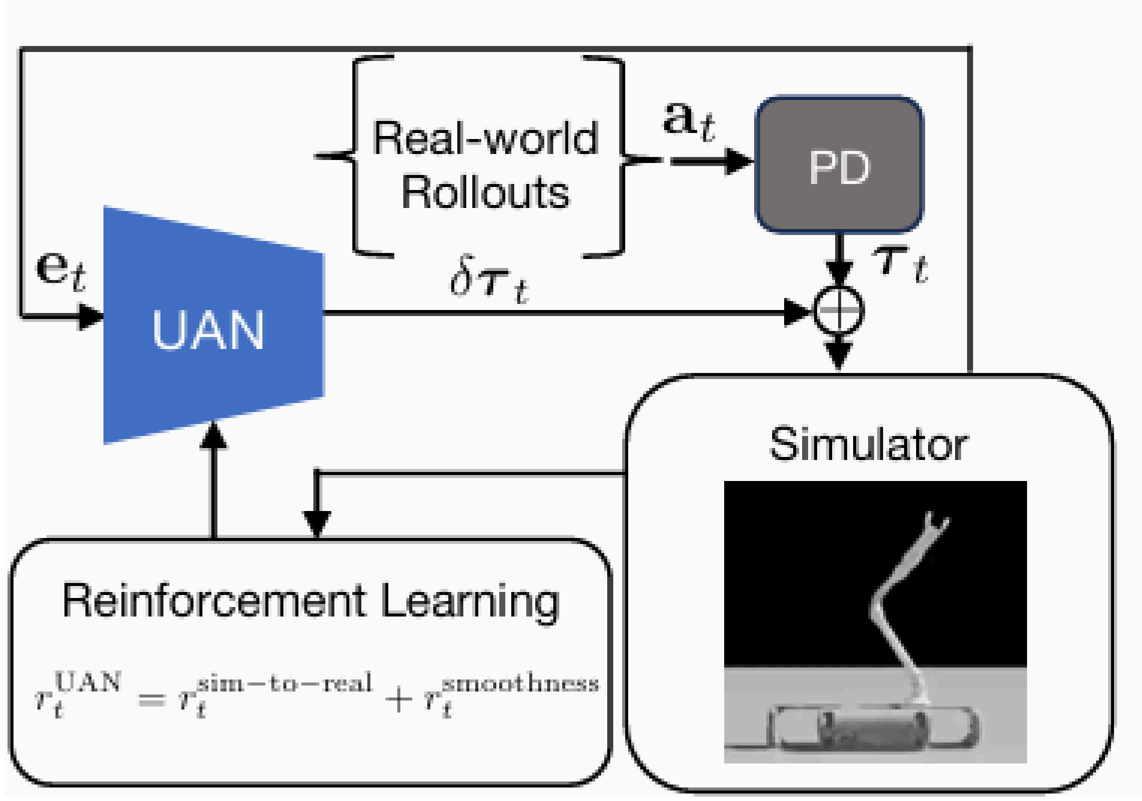
Sled Pull



Focus on this

Unsupervised Actuator Net (UAN)

- What is UAN?
  - A lightweight neural network acting as a "physics corrector" embedded directly within the simulation loop.
  - It maps a history of joint position and velocity errors to corrective torques.
- Goal: Correcting simulator dynamics without torque sensors.



Result

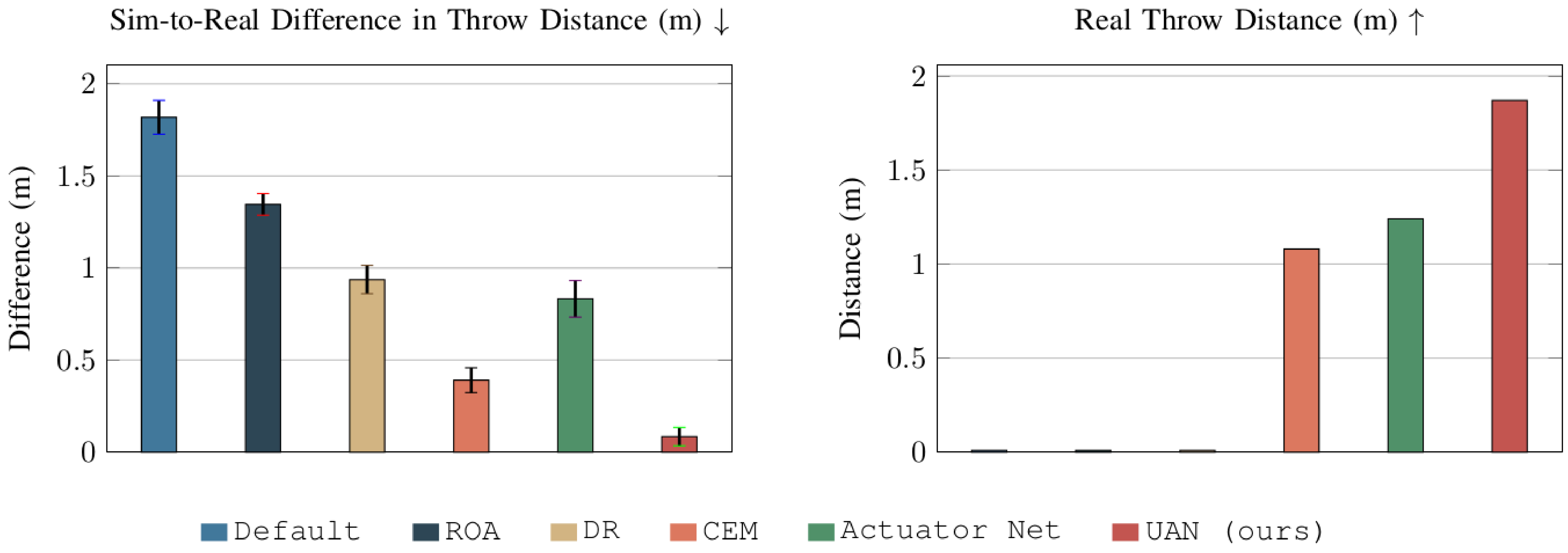


Fig. 4: **UAN improves simulator accuracy and real throwing performance.** `UAN (Ours)` achieves lower sim-to-real difference in throw distance as compared to standard baselines, resulting in a better real throw distance. For this comparison, we train and test policies with a fixed-base arm, to avoid the risk of the legged base falling during performance-critical ablations.

UAN

UAN Technique: Residual Actuator Correction

$$x_t = [q_t^{cmd} - q_t^{sim},\ q_t^{sim},\ \dot{q}_t^{sim},\ q_t^{cmd} - q_{t-1}^{cmd},\ a_{t-1}]$$

- Per-step dimension: $3 + 3 + 3 + 3 + 3 = 15$.
- History length H gives observation size $15H$.
- Default $H = 20$ gives 300 dimensions.

Why this is deployable

- Uses command, simulated state, and previous action.
- Does not require paired hardware state at runtime.
- Can be inserted into a future walking simulator.

UAN Technique: Residual Actuator Correction

Residual action

$$a_t = \pi_\theta(o_t), \qquad a_t \in [-1, 1]^3$$

$$\tau_t^{res} = 2.8 a_t$$

The recorded command remains the nominal target. UAN only adds residual effort.

Reward

$$r_t = 4.0 \exp\left(-\left\|\frac{q_{t+1}^{real} - q_{t+1}^{sim}}{0.05}\right\|^2\right) + 0.5 \exp\left(-\left\|\frac{\dot{q}_{t+1}^{real} - \dot{q}_{t+1}^{sim}}{0.5}\right\|^2\right) - 0.02\|a_t - a_{t-1}\|^2 - 0.001\|a_t\|^2$$

SIMULATION

Project

Category	Parameter	Value
Simulator	task	`oneleg-uan-deploy`
Simulator	timestep / decimation	0.005 s / 2
Robot	actuated joints	hip, knee, ankle
Training	parallel environments	4096
RL algorithm	PPO	RSL-RL
Policy	actor/critic MLP	[128, 128], ELU
Observation	per-step features	15
Observation	default history/dim	$H = 20$, 300 dim
Action	residual effort scale	2.8
Reward	q/dq weights	4.0 / 0.5

Dataset	Main characteristic	Purpose
walk-air	gait-like motion without contact	main position correction test
walk-contact	gait-like motion with foot contact	contact stress test
sine	smooth periodic command	clean training/evaluation signal
Gaussian	smooth random-like variation	state coverage and robustness
square	abrupt discontinuities	delay and transient stress case
chirp	changing frequency	speed-dependent response
triangle	ramp and reversal	direction-change response

Why this matters

A model trained only on smooth data may not handle abrupt or contact-rich motions.

EXPERIMENT

Table 6: Final deployable experiments used in this report.

Experiment	Evaluation	Main question
Exp1	walk-air, $H = 1, 5, 10, 20, 40$	How much history is useful with deployable observations?
Exp2	walk-air, 5–100% data subsets	How does data amount affect correction?
Exp3	walk-air, final H20	Does the updated final deployable policy reduce sim-to-real replay gap?
Exp4	OOD waveforms	Does sine-only deployable UAN generalize?
Exp5	walk-contact, final H20	Does the updated final policy remain useful under contact?

Position RMSE

$$\mathrm{RMSE}_q = \frac{1}{3}\sum_{j=1}^{3}\sqrt{\frac{1}{N}\sum_i (q_{i,j}^{real} - q_{i,j}^{sim})^2}$$

Velocity RMSE

$$\mathrm{RMSE}_{\dot{q}} = \frac{1}{3}\sum_{j=1}^{3}\sqrt{\frac{1}{N}\sum_i (\dot{q}_{i,j}^{real} - \dot{q}_{i,j}^{sim})^2}$$

Residual action RMS

$$\mathrm{RMS}_a = \frac{1}{3}\sum_{j=1}^{3}\sqrt{\frac{1}{N}\sum_i a_{i,j}^2}$$

Interpretation rule

q RMSE is the main tracking metric. dq RMSE and action RMS show the cost and dynamic quality of the correction.

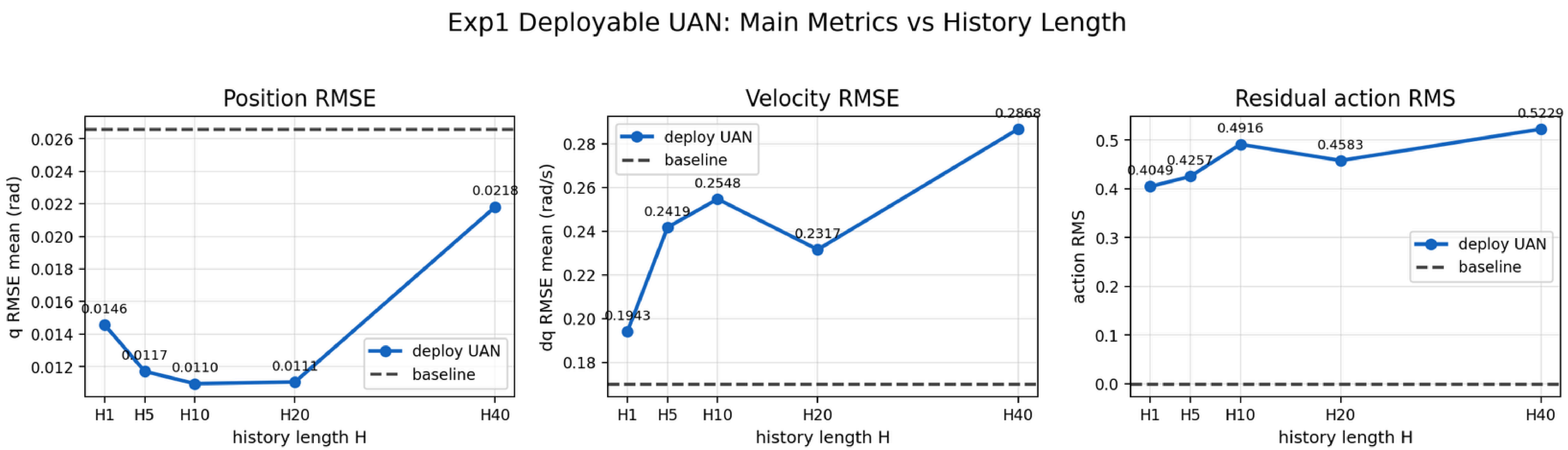
- **q RMSE**: main performance metric for joint-position replay.
- **dq RMSE**: velocity replay error; useful but sensitive to encoder differentiation and filtering.
- **action RMS**: residual effort magnitude; not a tracking metric by itself.
- **baseline action RMS $= 0$**: because the baseline applies no residual action.

Main interpretation rule

A good UAN should reduce q RMSE while keeping dq RMSE and residual action reasonably low.

Exp1 Result: History Length Ablation

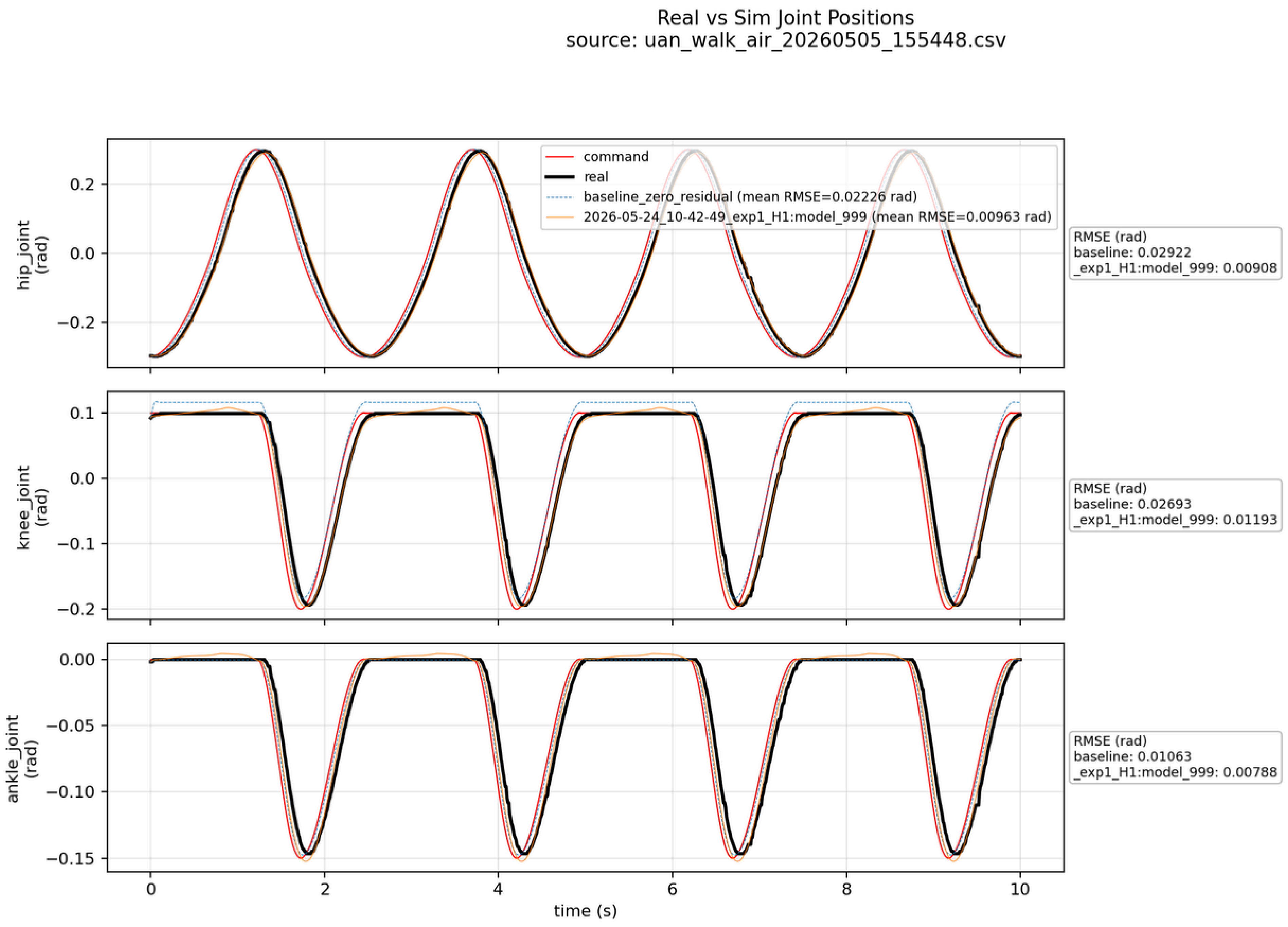
H	q RMSE	dq RMSE	Action
1	0.01457	0.19427	0.40489
5	0.01172	0.24187	0.42567
10	0.01096	0.25480	0.49159
20	0.01106	0.23169	0.45826
40	0.02181	0.28682	0.52287

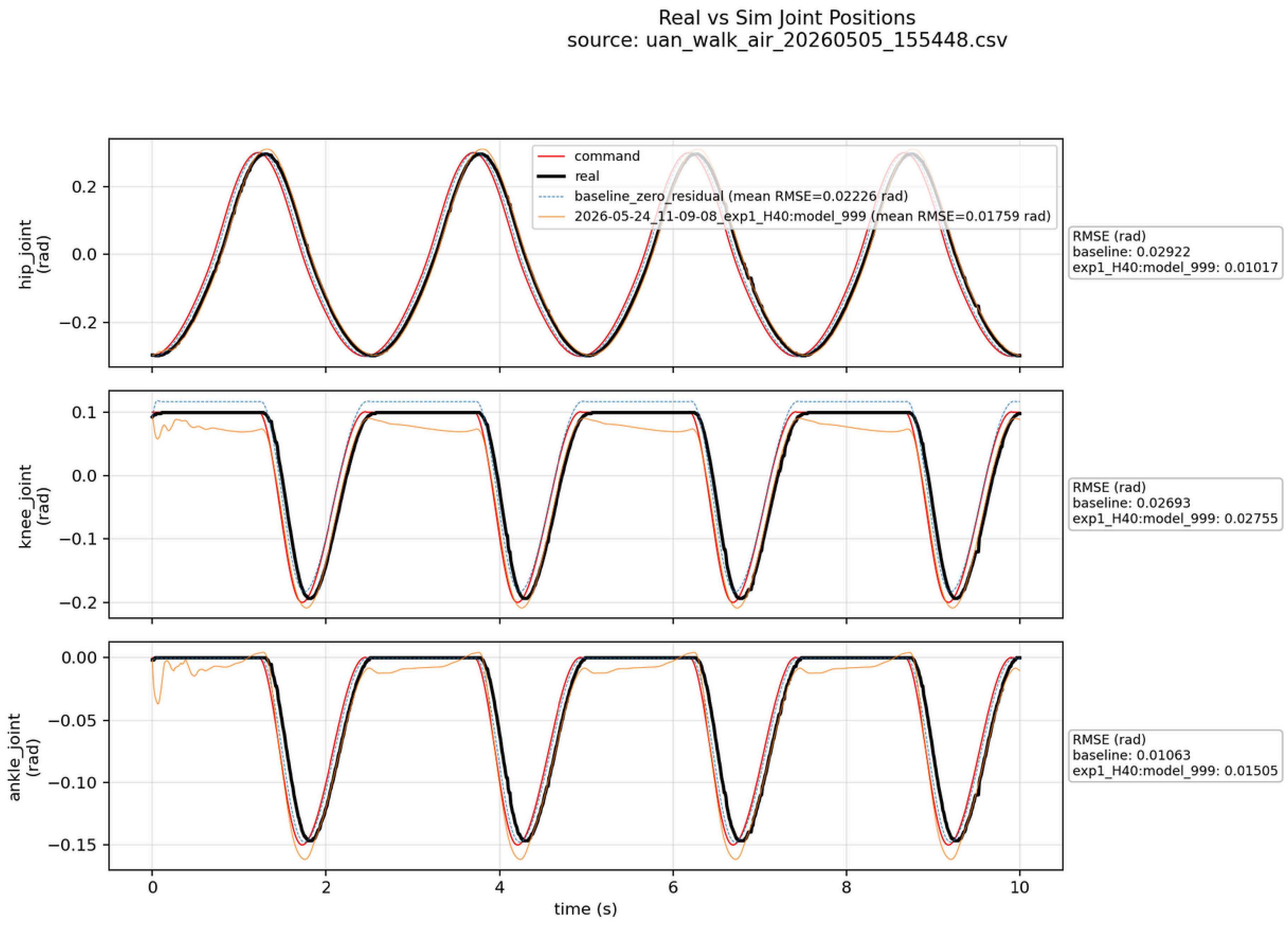


Takeaway

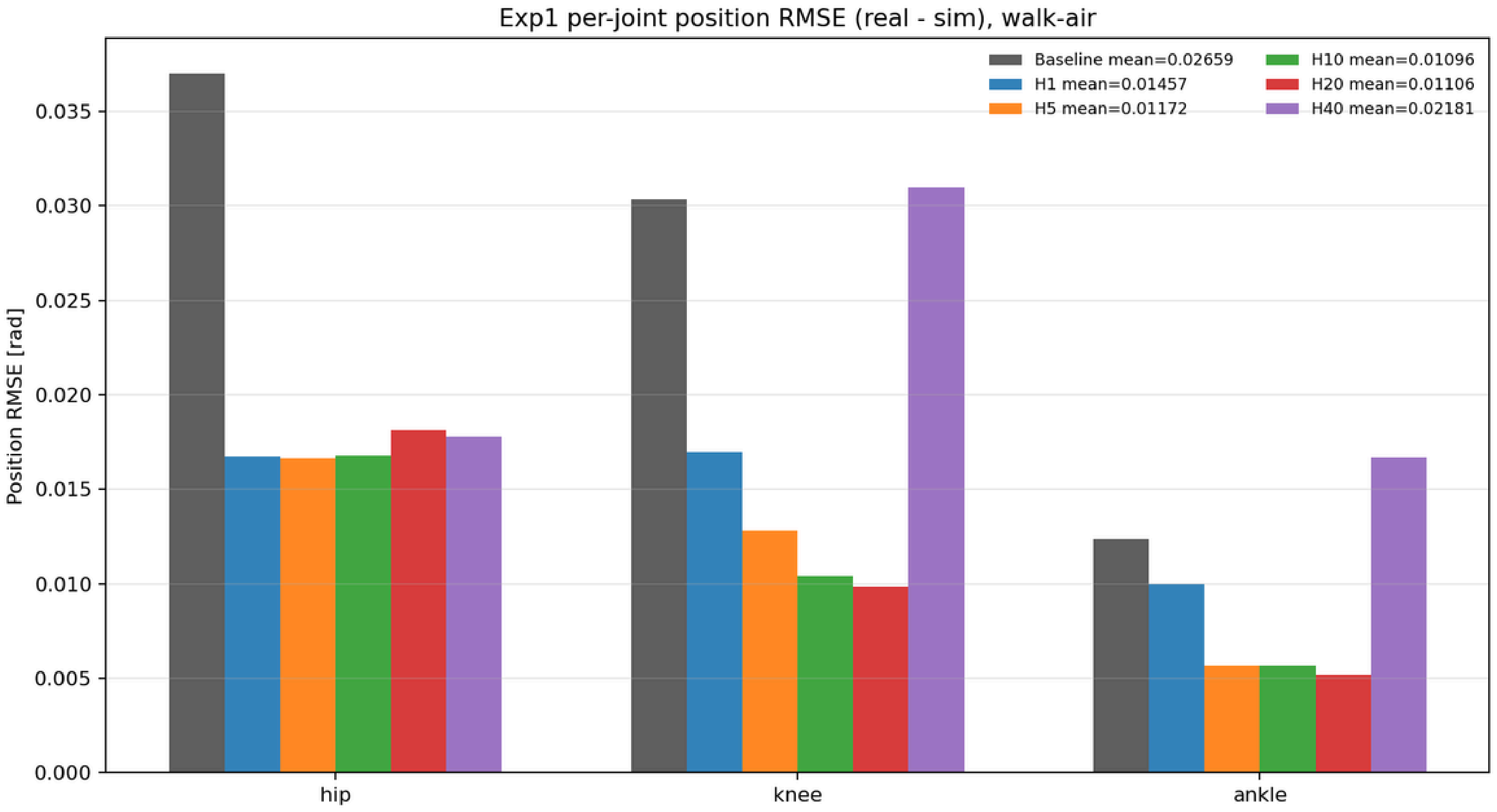
H10/H20 are the best trade-off region. H40 becomes worse because the observation becomes too large and stale.

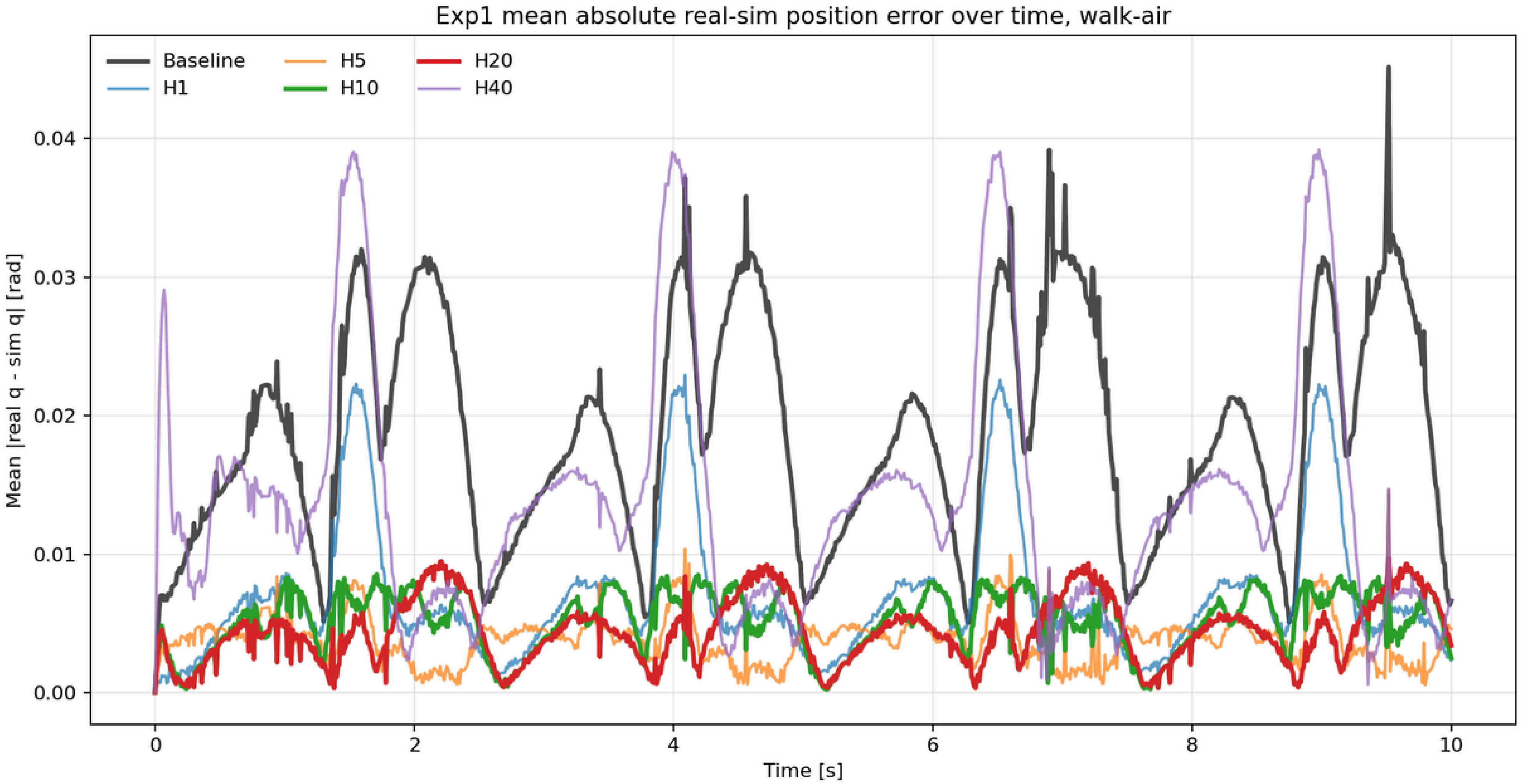
Exp1 Result: History Length Ablation





Exp1 Result: History Length Ablation



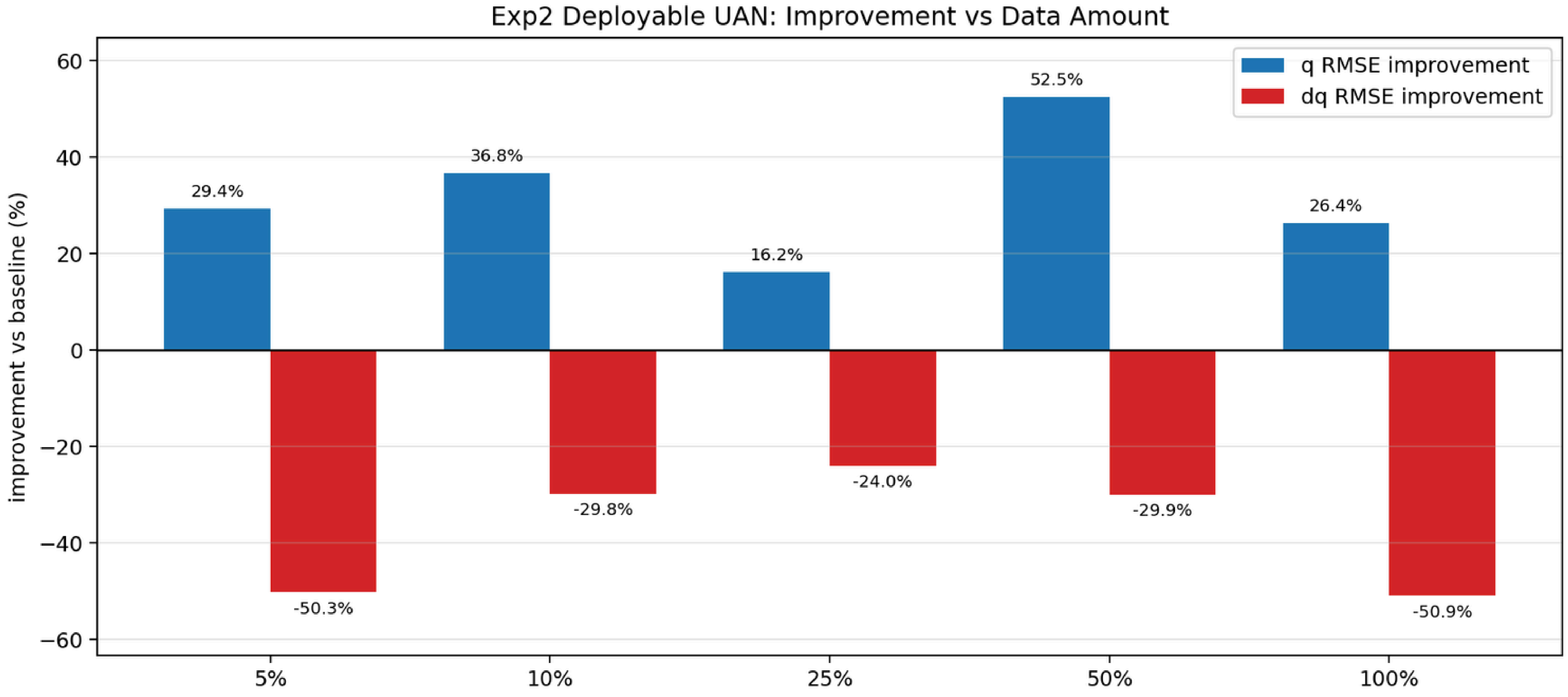


Exp2: Data and Sample Efficiency

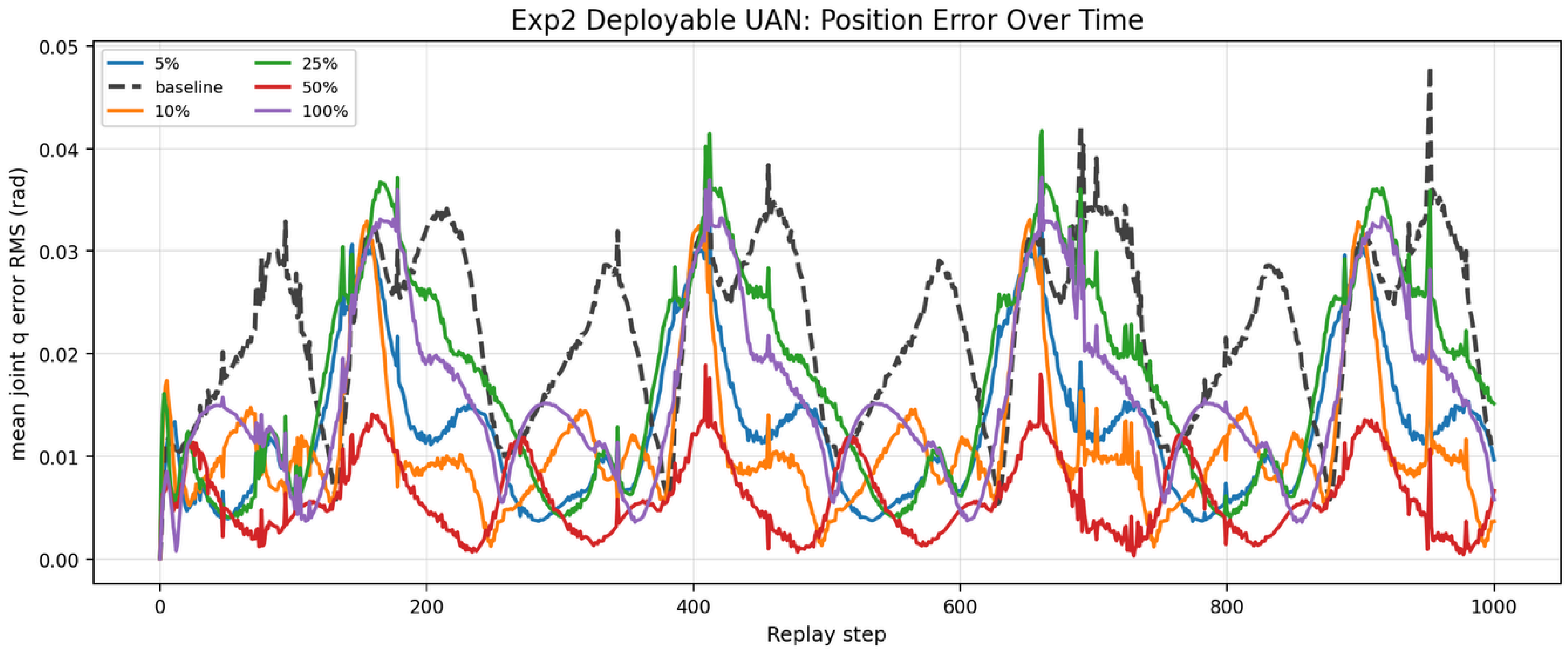
Data	q RMSE	Improve	Action
5%	0.01877	29.4%	0.40067
10%	0.01681	36.8%	0.41964
25%	0.02227	16.2%	0.33827
50%	0.01264	52.5%	0.42421
100%	0.01958	26.4%	0.37053

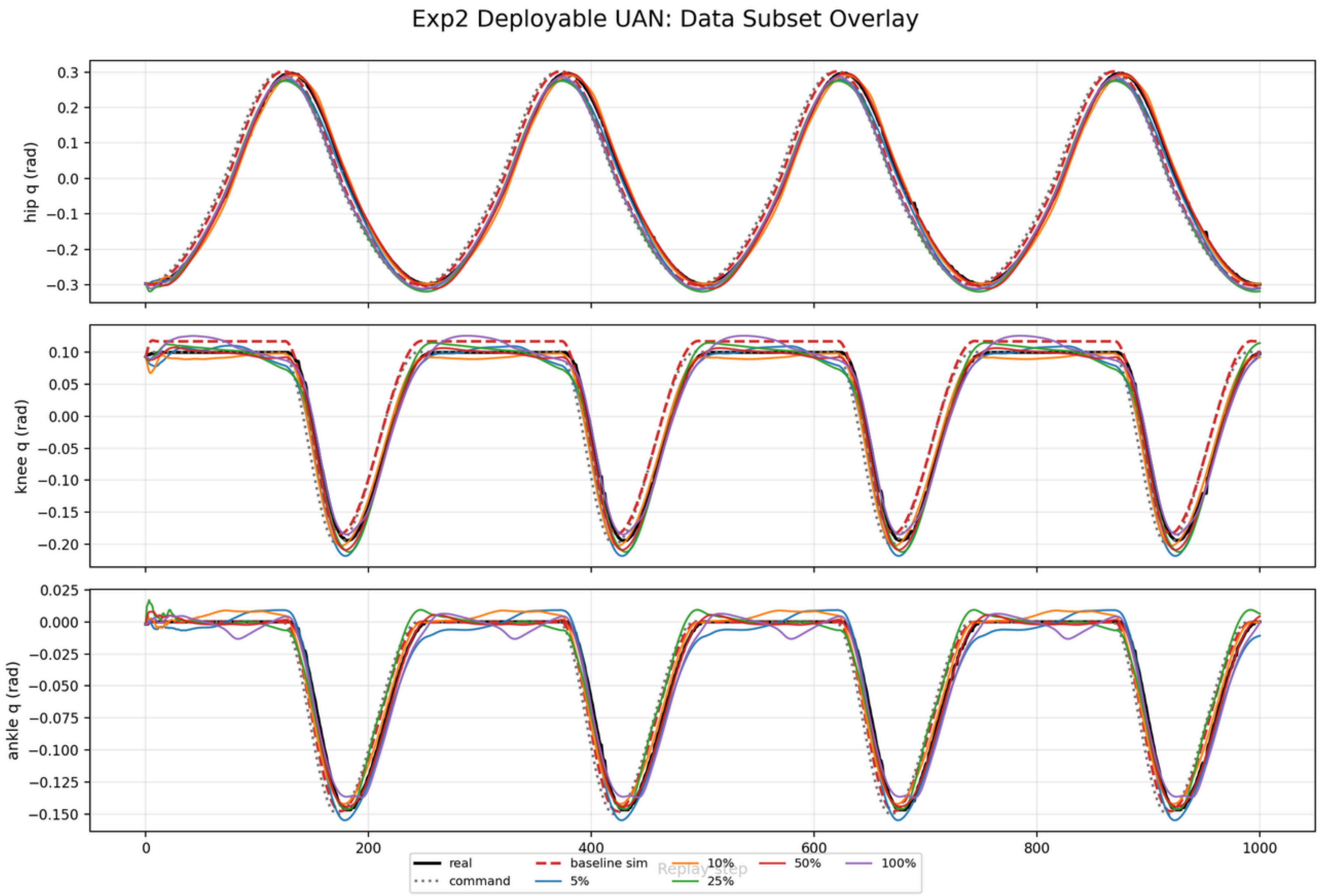
Key point

More data is not automatically better in this run. p50 is best; p100 is weaker.



Exp2: Data and Sample Efficiency



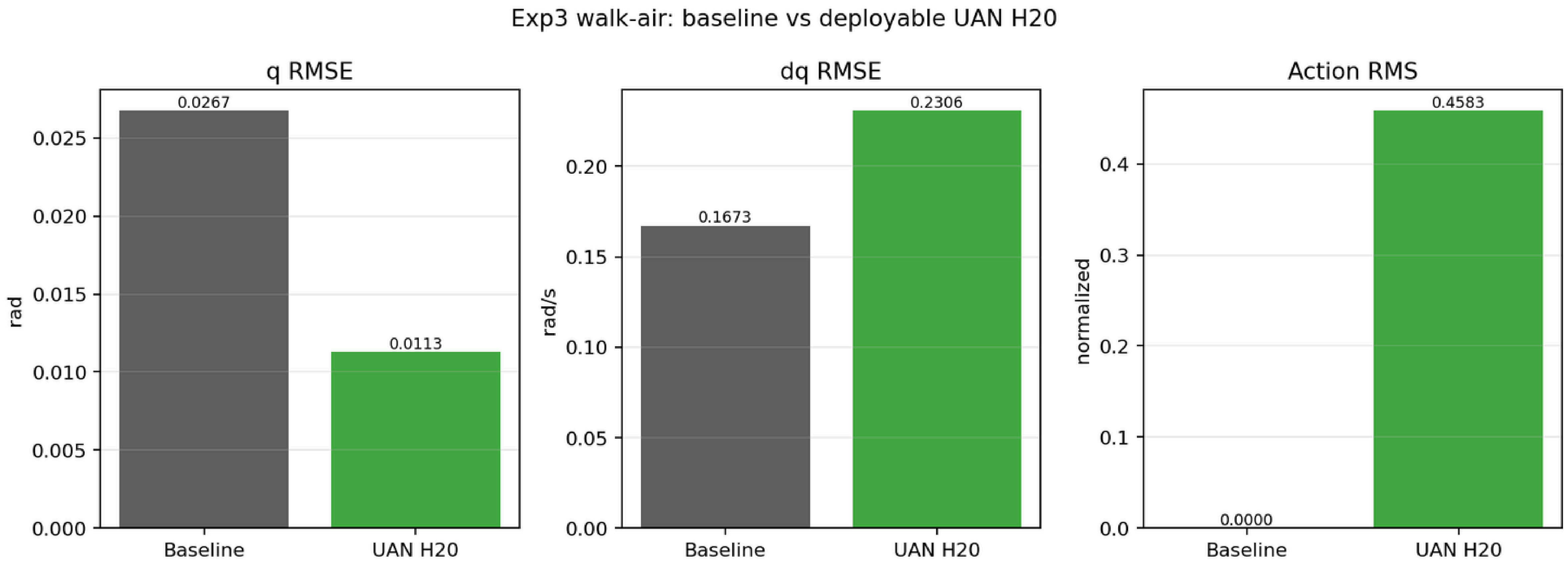


Exp3: Walk-Air Sim-to-Real Gap

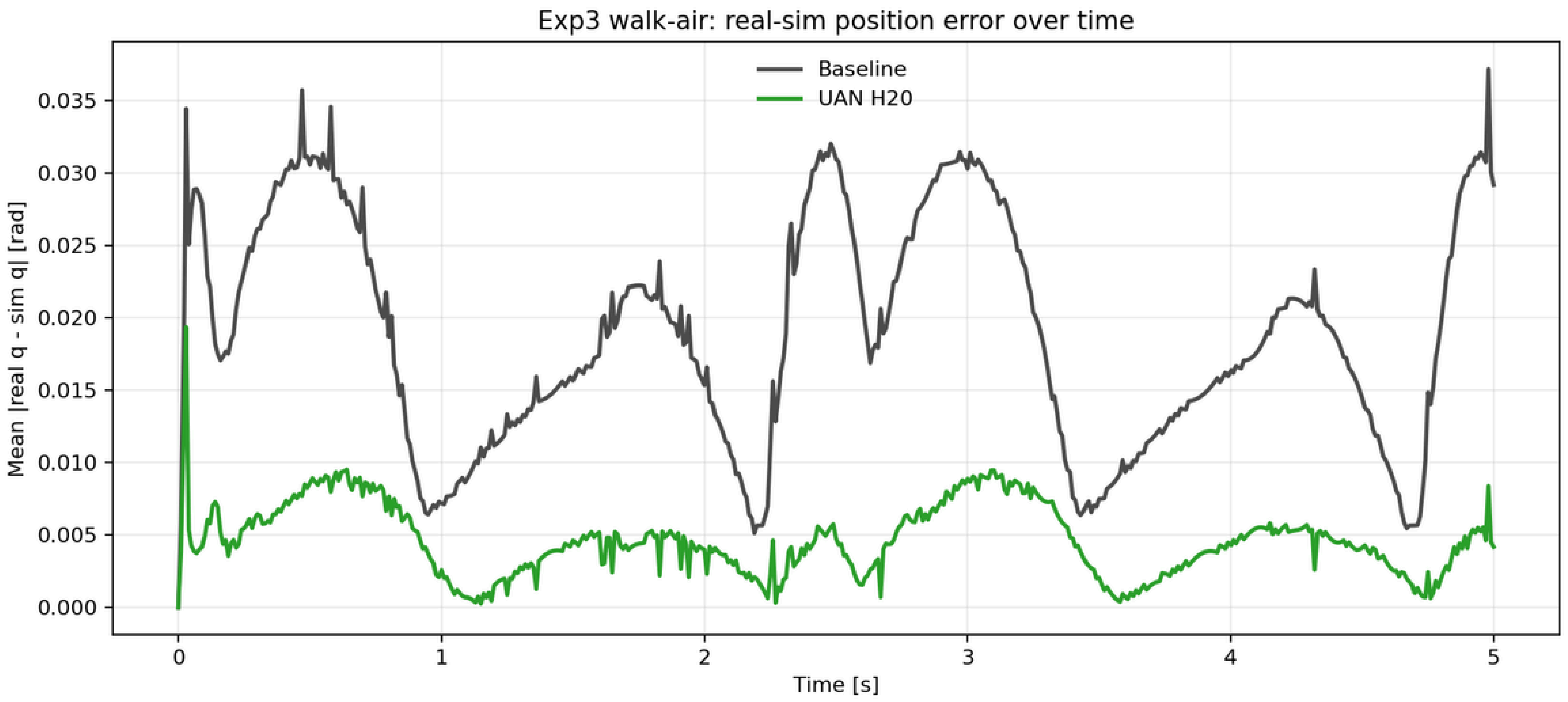
Case	Base q	UAN q	Improve	Action
H20 final	0.02673	0.01129	57.8%	0.45827

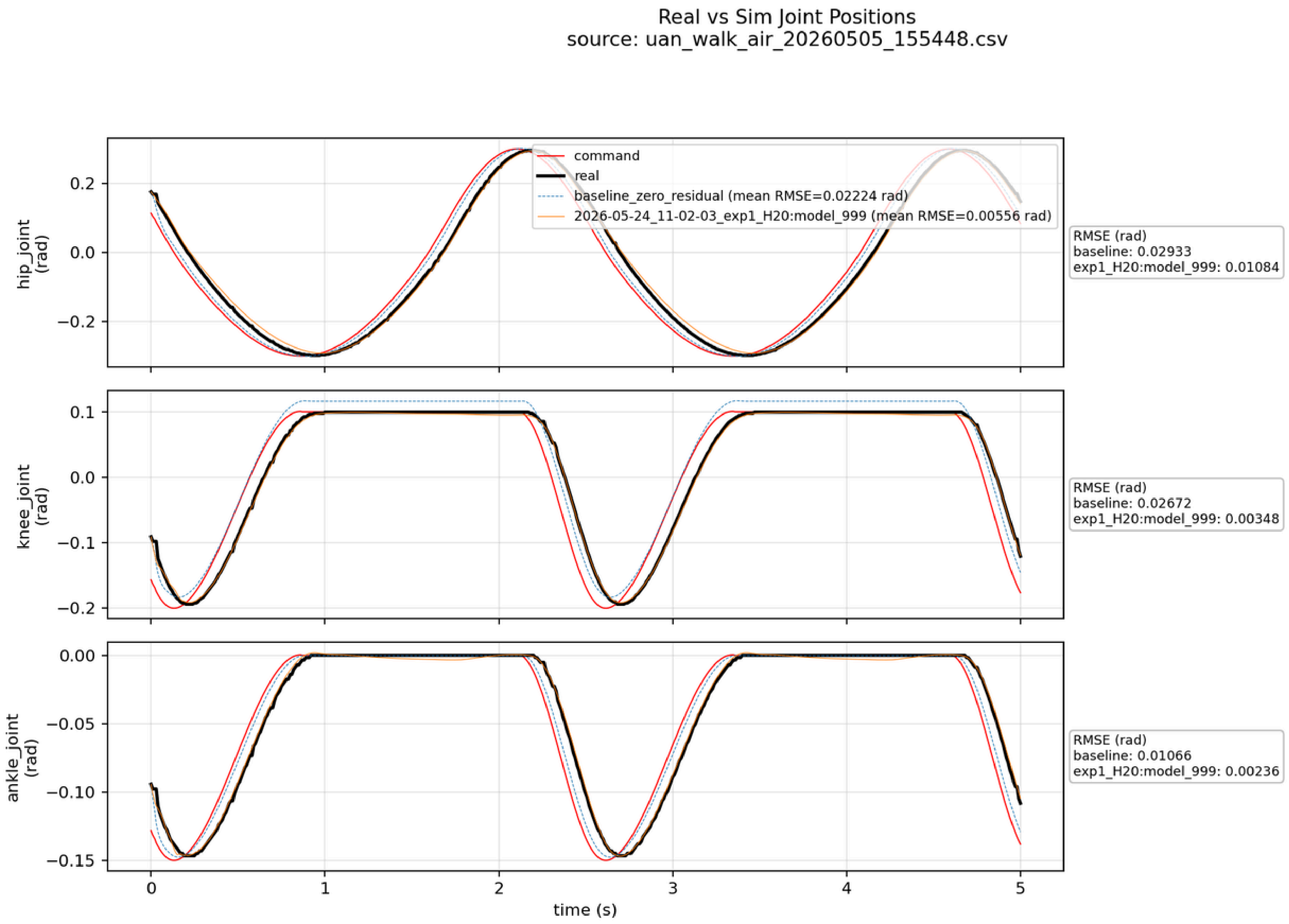
Takeaway

The updated final policy strongly improves walk-air position replay, while dq RMSE still increases.



Exp3: Walk-Air Sim-to-Real Gap



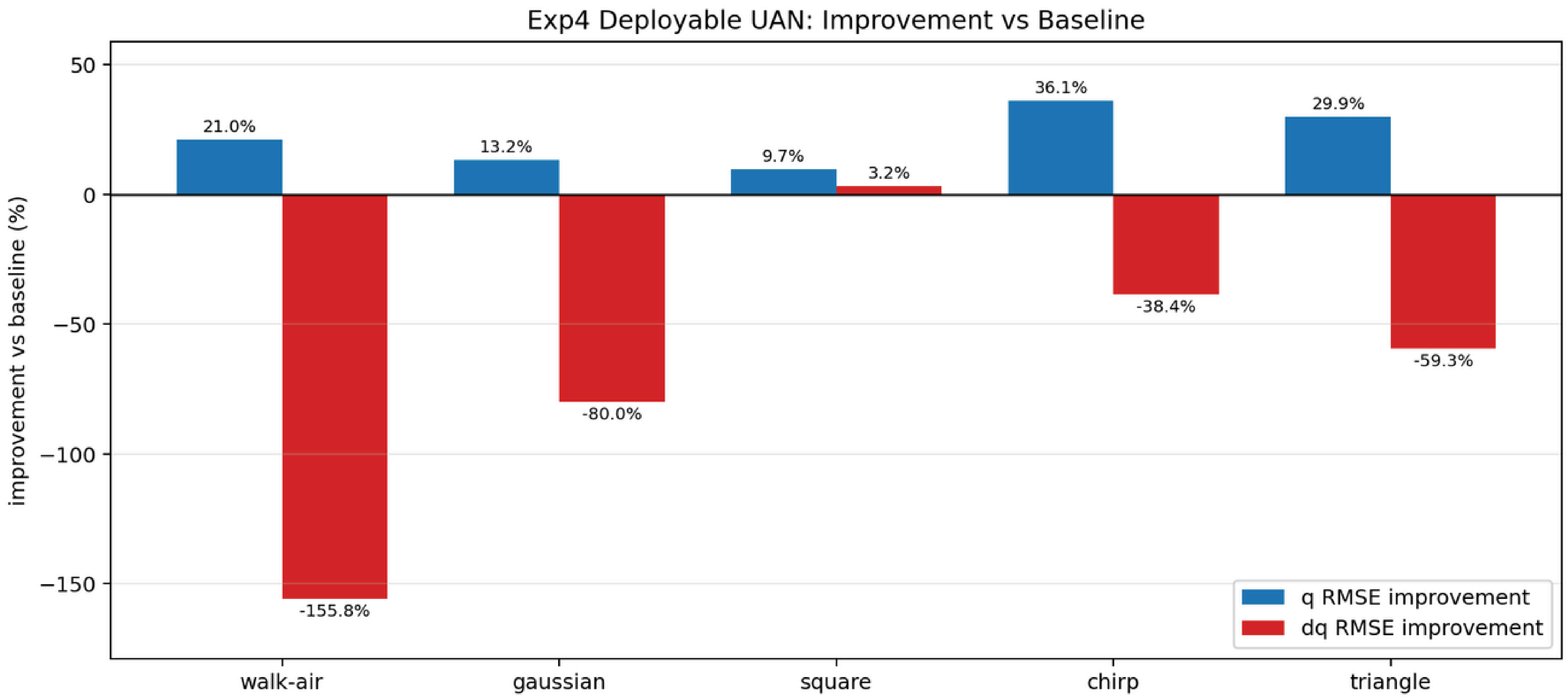


Exp4: Out-of-Distribution Generalization

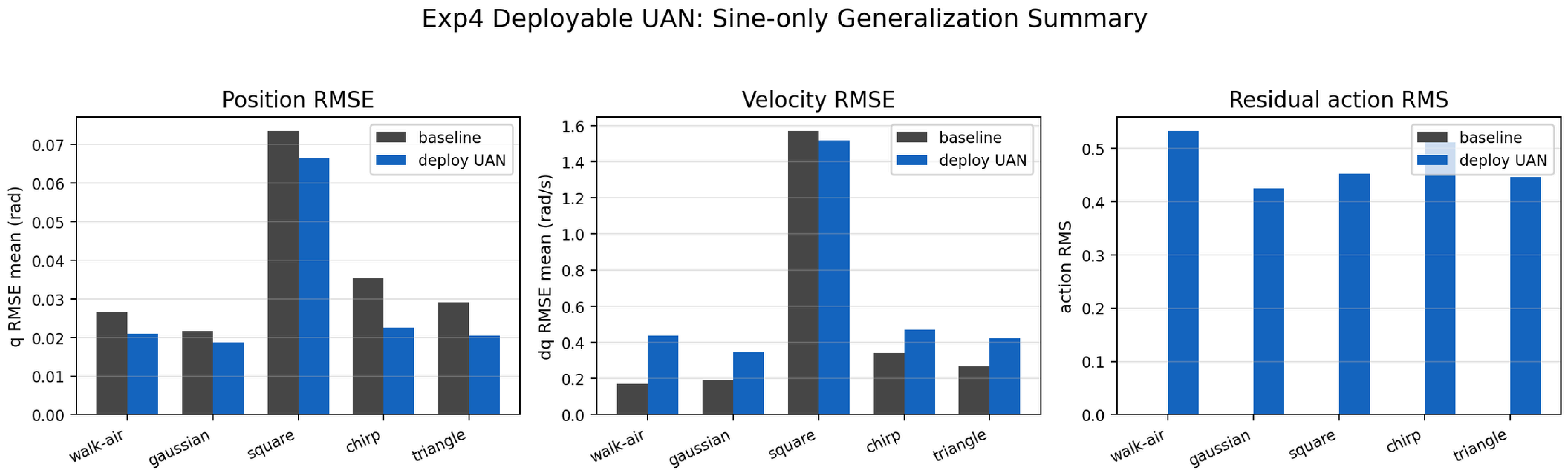
Data	Base q	UAN q	Improve
walk-air	0.02659	0.02099	21.0%
Gaussian	0.02169	0.01883	13.2%
Square	0.07351	0.06637	9.7%
Chirp	0.03533	0.02259	36.1%
Triangle	0.02911	0.02040	29.9%

Takeaway

The sine-only model improves q RMSE on all datasets, but square remains hardest.



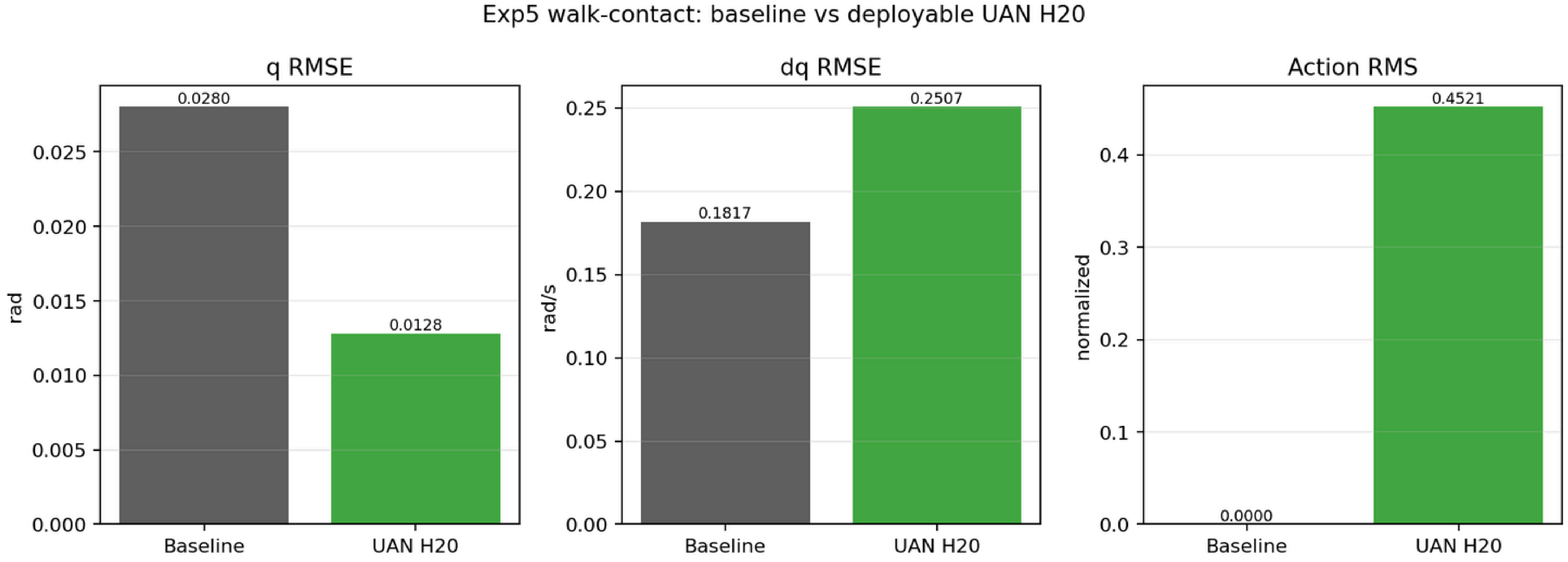
Exp4 Result: Sine-Only Generalization



Case	Base q	UAN q	Improve	Action
H20 final	0.02803	0.01281	54.3%	0.45209

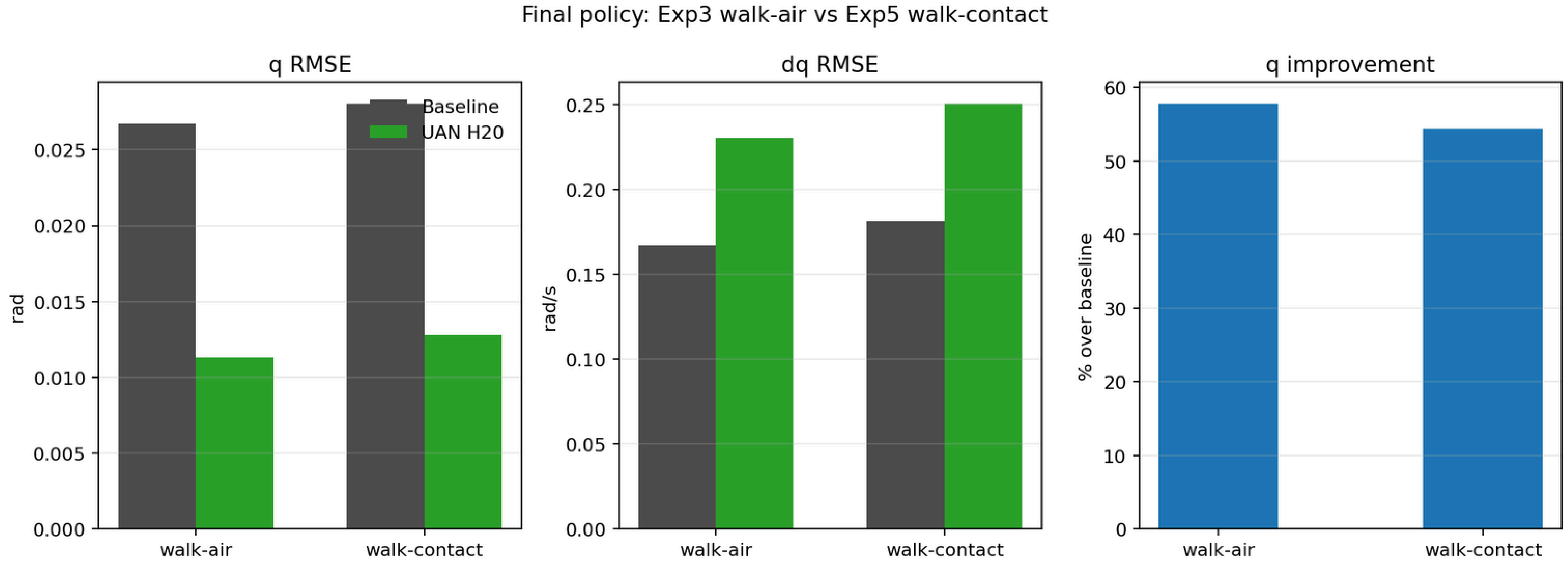
Takeaway

Contact q RMSE improves, but dq RMSE remains worse than baseline.



Overall Walk-Air vs Walk-Contact Summary

- Final walk-air: q RMSE 0.02673 → 0.01129.
- Final walk-contact: q RMSE 0.02803 → 0.01281.
- Position improvement is consistent.
- Velocity RMSE usually increases.
- Residual effort is nonzero and must be reported.



Why position improves

- Reward strongly emphasizes q tracking.
- Residual effort compensates actuator lag/friction.
- Command error and simulator state are enough for useful correction.

Why velocity worsens

- Velocity labels may be noisy or filtered.
- dq reward weight is smaller.
- Aggressive residual action can improve next position but hurt smoothness.

Safe interpretation

Current deployable UAN is a successful actuator position corrector, not yet a complete dynamic/contact model.

- In Exp2, p50 has enough useful transition diversity to learn strong correction.
- p100 may include more heterogeneous, noisy, or conflicting transitions.
- A fixed training budget may be insufficient for the full dataset.
- PPO stochasticity and subset composition matter.

Report conclusion for Exp2

Exp2 is evidence of data sensitivity, not a final scaling law. A stricter study needs multiple seeds and a frozen hold-out set.

- Velocity RMSE is worse in most experiments.
- Residual action RMS is nonzero and sometimes large.
- Exp2 is non-monotonic and was not repeated across random seeds.
- Contact test is replay evaluation, not full payload robustness.
- One-leg actuator calibration is not yet full hexapod locomotion.

CONCLUSION

What we built

A deployable UAN actuator-correction pipeline for a 3-DOF one-leg rig.

What worked

The final walk-air model reduces q RMSE from 0.02673 to 0.01129, and the final walk-contact model reduces q RMSE from 0.02803 to 0.01281.

What remains

Velocity dynamics, residual smoothness, stricter validation, and full hexapod integration.

Final takeaway: deployable UAN is a strong proof of concept for actuator position correction.

THANK YOU